'his question paper contains 8 printed pages.]

Your Roll No. 30/05/2023

. No. of Question Paper : 4842 **E**

nique Paper Code : 32517916

ame of the Paper : Embedded Systems

ame of the Course : **B.Sc. (H) Electronics (DSE)**

emester : VI (CBCS)

uration : 3 Hours Maximum Marks : 75

structions for Candidates

Write your Roll No. on the top immediately on receipt of this question paper.

There are **seven** questions in all, out of which you have to attempt any **Five** questions.

Q1. is compulsory. **All** Questions carry equal marks.

Use of non-programmable scientific calculator is allowed.

(a) Differentiate between RISC and CISC architecture. (3)

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(b) Explain the difference between the S, N and V flags and where each one is used? (3)

(c) Show how to enable all three external hardware interrupts. (3)

(d) What is the baud rate if we use UBBR = 15 to program the baud rate? Assume XTAL = 10MHz. (3)

(e) How does the SPI protocol distinguish between the read and write cycles? (3)

2. (a) Briefly explain five major criterion for choosing a microcontroller for an embedded system. (5)

(b) Find the address of the last location of on chip program ROM for each of the 5 following :

AVR with 8 KB

AVR with 128 KB (5)

(c) Illustrate the difference between the following instructions using examples : (5)

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(i) IN and LDS

(ii) SBIS and SBRS

(a) What is a Stack and what is its role when a subroutine is called? Upon Reset, Stack Pointer points to which location? Explain working of the instructions that are used to store and retrieve data from Stack? (5)

(b) What is an instruction cycle time for an AVR? Find the period of the instruction cycle when the clock frequency is 16MHz. Also, explain branch penalty. When does it occur? (5)

(c) A tap sensor is connected to bit 1 of Port B and represents the condition of the 5 sensor. If the sensor goes LOW, it means that it is activated. Monitor the sensor bit continuously. An SCR is connected to bit 7 of Port C.

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Write an AVR program to monitor the tap sensor and when hands are brought close to the tap, the sensor gets activated, it sends a HIGH to turn on the SCR and makes water flow out of the tap. (5)

4. (a) Write a program that finds the number of zeroes in an 8 bit number. (5)

(b) What are the four major reasons for writing programs in C instead of Assembly? Also, enumerate any disadvantage of C over assembly, if any. (5)

(c) What is the value of R20 after the execution of the following code? (5)

```
LDI R20,0x24

SEC

ROL R20

CLC
```

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LSR R20

SEC

ASR R20

CLC

LSL R20

SEC

ROR R20

3. (a) What are the different ways in which Atmega32 can be reset? Explain. The RESET is normally _____ (LOW, HIGH) and needs a _____ (LOW, HIGH) signal to be activated. (5)

(b) What are the various sources of Interrupt? Give at least four. What are the two ways in which a microcontroller can execute a service request made by a device? Compare both the methods. (5)

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(c) Assume that Timer0 is programmed in CTC mode, OCR0 = 0X30 and the compare match interrupt is enabled. Explain how the interrupt for the timer works. Assume XTAL = 8 MHz, and the timer is in no prescaler mode. How often the ISR is executed? (5)

6. (a) Using a prescaler of 64, write a program to generate a delay of 1520μs. Assume XTAL = 8 MHz, Normal mode. (5)

(b) Assume that XTAL = 1 MHz. Find the OCR0 value needed to generate a time delay of 0.2 ms. Use CTC mode and no prescaler. Also determine which all options of prescaler can be used to generate the same time delay. (5)

(c) Write a program to receive bytes of data serially asynchronously and put them on Port C. Set the baud rate at 19200, 7 bit data, even parity and 2 stop bits. (5)

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7. (a) How do we start Analog to Digital conversion in the ATmega32? Also, how is the End of conversion recognized in ATmega32. Find the step size for a 10 bit AVR ADC. Also, calculate the first conversion time if F_{osc} = 4 MHz, ADPSO-2 = 110 (Clk/64) (5)

(b) Why REPEATED START condition is necessary in I2C bus protocol? What will happen if the master does not use it?

Explain the data packet format for an I2C protocol. (5)

(c) Compare SPI protocol with the I2C communication protocol In accessing SPI devices, explain single byte and multibyte burst mode of operation. (5)

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Appendix

GICR	INT1	INT0	INT2				IVSEL	IVCE
GIFR	INTF1	INTF0	INTF2					
TIMSK	OCIE2	TOIE2	TICIE1	OCIE1A	OCIE1B	TOIE1	OCIE0	TOIE0
TIFR	OCF2	TOV2	ICF1	OCF1A	OCF1B	TOV1	OCF0	TOV0
UCSRA	RXC	TXC	UDRE	FE	DOR	PE	U2X	MPCM
UCSRB	RXCIE	TXCIE	UDRIE	RXEN	TXEN	UCSZ2	RXB8	TXB8
UCSRC	URSEL	UMSEL	UPM1	UPM0	USBS	UCSZ1	UCSZ0	UCPOL
TCCR0	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
MCUCR	SE	SM2	SM1	SM0	ISC11	ISC10	ISC01	ISC00
ADCSRA	ADEN	ADSC	ADATE	ADIF	ADIE	ADPS2	ADPS1	ADPS0
ADMUX	REFS1	REFS0	ADLAR	MUX4	MUX3	MUX2	MUX1	MUX0

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